TCX2005: Information Systems, Management and Organisations

L5: IS Posed Challenges & Issues

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Lesson Objectives

- Addressing challenges posed by strategic IS
- Ethical and Social Issues raised by IS
- Legal Issues in IS

Learning Resources

- Textbook (read before class)
  - Read chapter 3, 4
- Any Other Material
  - Shared on Canvas for this week

Case Study 01

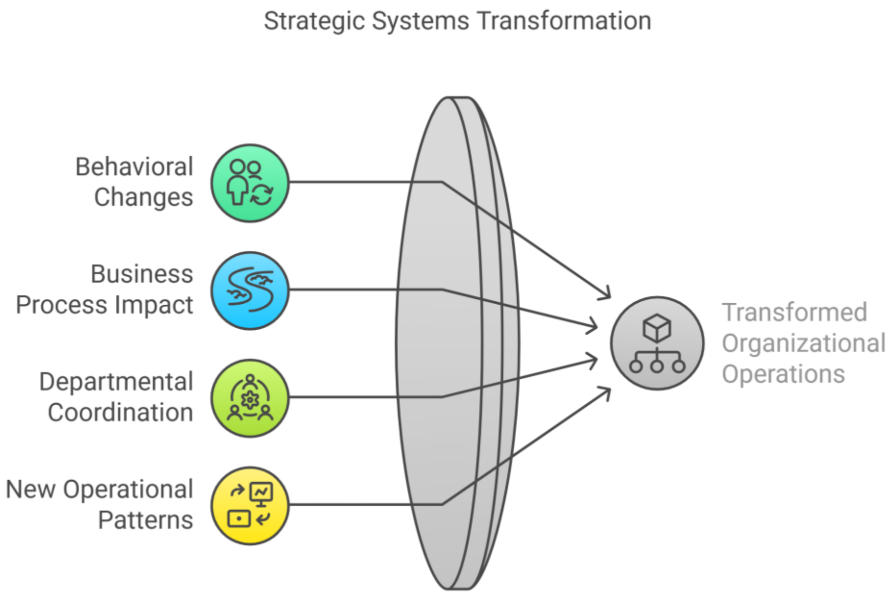
Read the case study from textbook on -Sharp Organization (revisit same from Lecture-2)

Points to understand:

- Implemented Yammer for internal communication in 2013
- The Sharp case perfectly illustrates what we call "socio-technical change" - where technology and social/organizational changes are intertwined.
  - Sharp didn't just install new software; they fundamentally changed how people worked and communicated.
  - Cultural shift in a traditional Japanese company
- Analyzing this case reveals typical IS implementation challenges:
  1. Potential resistance from hierarchical business cultures
  2. Adoption hurdles for new communication technologies
  3. Organizational learning requirements for effective use

The Nature of Strategic IS Challenges

- Strategic Information Systems fundamentally transform how organizations operate. Unlike routine operational systems, they create deep organizational changes that present unique challenges.
  - Changes organizational behavior
    - E.g., at Sharp, employees shifted from formal hierarchical communication to open, cross-level dialogue on Yammer
  - Affects all business processes
    - E.g., NUS's switch to Canvas affected how students learn, teachers teach, and staff administrate
  - Requires coordination across departments
    - E.g., ComfortDelGro's transition to digital booking required alignment between IT, Operations, Customer Service and finance department.
  - Creates new operational patterns
    - E.g., Sharp's product development became more collaborative with inter-department feedback loops

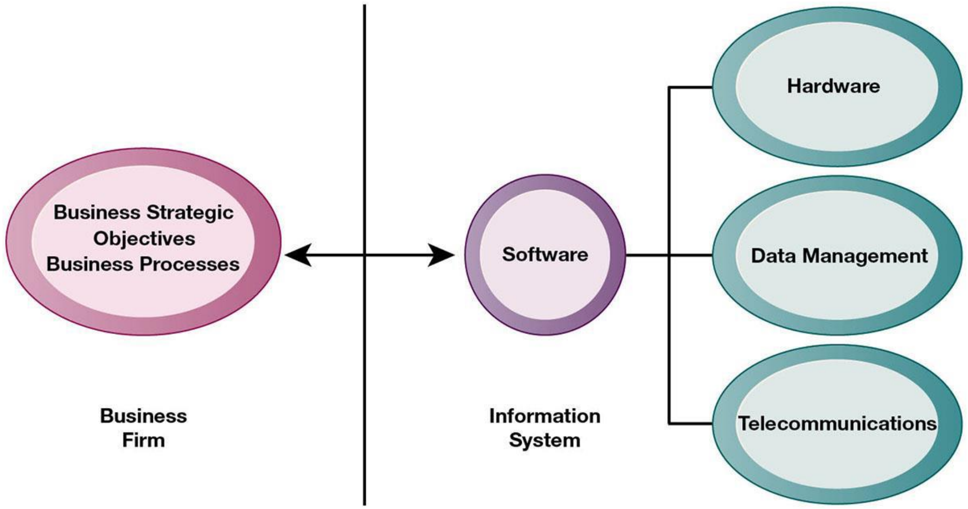


The Competitive Advantage Paradox

- Strategic advantages from IS are increasingly temporary.
- Example: The evolution of e-commerce
    - 1995: Amazon revolutionizes online shopping as a first mover
    - 2000s: Walmart, Target launch competing platforms
    - 2020s: E-commerce becomes a basic requirement for survival
- Good ideas get copied fast!
- Customers expect more ➔ Organization need to keep innovating

Business-IT Alignment Challenge

- Despite billions spent on IT systems, most organizations struggle to align their technology investments with business goals.
  - Only 25% of firms achieve proper alignment
  - Up to 50% of profits linked to IT alignment
  - Most companies struggle with integration (Luftman 2003)



Example: Impact of Delayed Business-IT Alignment

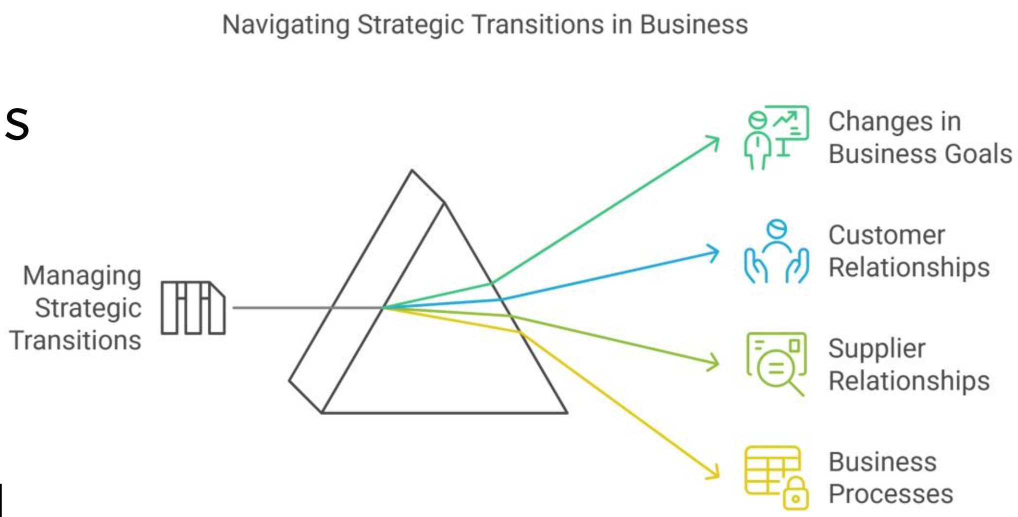
- ComfortDelGro's slow digital response to market disruption
  - Grab starts business in SG: 2013
  - Grab merges with Uber in 2018
  - ComfortDelGro trial ride-hailing service with private-hire cars from Feb 4, 2021
  - ComfortDelGro to launch all-in-one app (i.e., Zig) offering taxi, private bus, car rental and leasing services in 2021.
  - The collaboration between ComfortDelGro and Gojek in April, 2024

Task: Comment on ComfortDelGro Taxi's current Business/IT alignment in SG.

Managing Strategic Transitions

When organizations implement strategic IS, they face a fundamental challenge: *transforming their entire business while keeping it running*.

- Adopting strategic systems requires changes in:
    - Business goals,
        - E.g, ComfortDelGro: from maintaining traditional market share to competing in the digital mobility ecosystem
    - Relationships with customers and suppliers, and
    - Business processes
        - E.g., retraining staff and adapting operations
- Success, depends on oganizational culture readiness



Strategic Solutions

- Active Management Participation: Successful digital transformation requires leaders to drive technology strategy
  - E.g., Microsoft's Digital Transformation
  - CEO Satya Nadella leading cloud-first strategy
  - Now, integration of AI across product lines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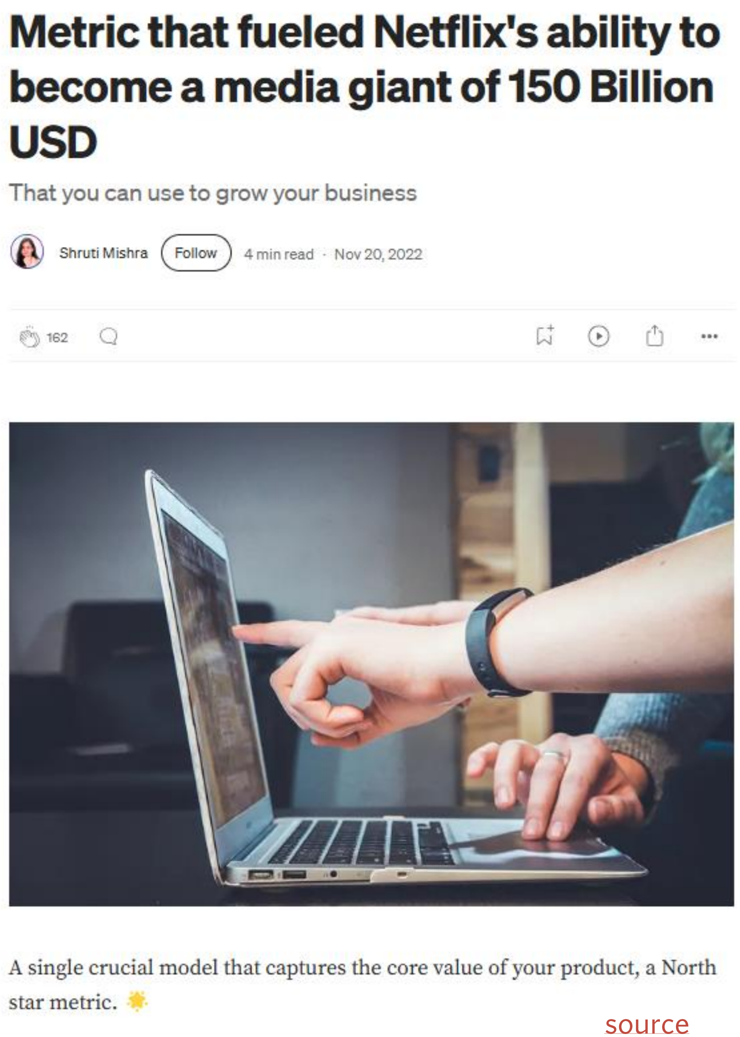
Strategic Solutions (cont.)

- Measuring IT Impact: “*you can’t manage what you can’t measure*”
    - Without clear metrics, organizations cannot tell if their IT investments are succeeding or failing.
    - E.g., for a successful data-driven decision Making at Netflix, they need to have proper performance measurements in place.



Metric that fueled Netflix's ability to become a media giant of 150 Billion USD

That you can use to grow your business

Shruti Mishra · Follow · 4 min read · Nov 20, 2022

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A single crucial model that captures the core value of your product, a North star metric.

source

Case Study 02

Read the case study from textbook on –Your Smartphone: Big Brother's Best Friend:

Identify:

a. The problem
b. Outcomes
c. The solution
d. Key learning point(s)

Note: be ready to share your thoughts

Case Study 02

Problem:

- Companies are extensively tracking and collecting location data from millions of cell phone users, often without their full awareness or informed consent
  - Limited regulation on how companies can use/share this data once collected
  - Difficult for users to know which companies have their data and how it's being used
  - Data claimed to be "anonymous" can still identify individuals through comprehensive location patterns
  - Complex privacy policies make it hard for average users to understand what they're agreeing to

Outcomes:

- Widespread collection and commercialization of user location data
- Creation of massive data repositories tracking people's movements
- Companies benefiting financially from selling location data
- Potential negative impacts on individuals including:
  - Privacy invasion and possible job discrimination
  - Unwanted targeted advertising
  - Exposure to security risks

Case Study 02

Solution:

- Some apps providing clearer disclosures about location tracking
- Apps revising consent screens to include more details (e.g., iHeartRadio's example)
- Users having the option to "consent" to data collection, though this is often required to use the app

(we also discussed many similar cases and solutions in TCX1003)

Learning:

- IS can be a "double-edged sword" - providing benefits while creating new risks
- Current privacy protections for mobile location tracking are inadequate
- The importance of informed consent and transparent data practices
- Managers need to balance the benefits of data collection against ethical responsibilities to users

Ethical and Social Issues raised by IS

Ethics:

- Principles of right and wrong that individuals, acting as free moral agents, use to make choices to guide their behaviors

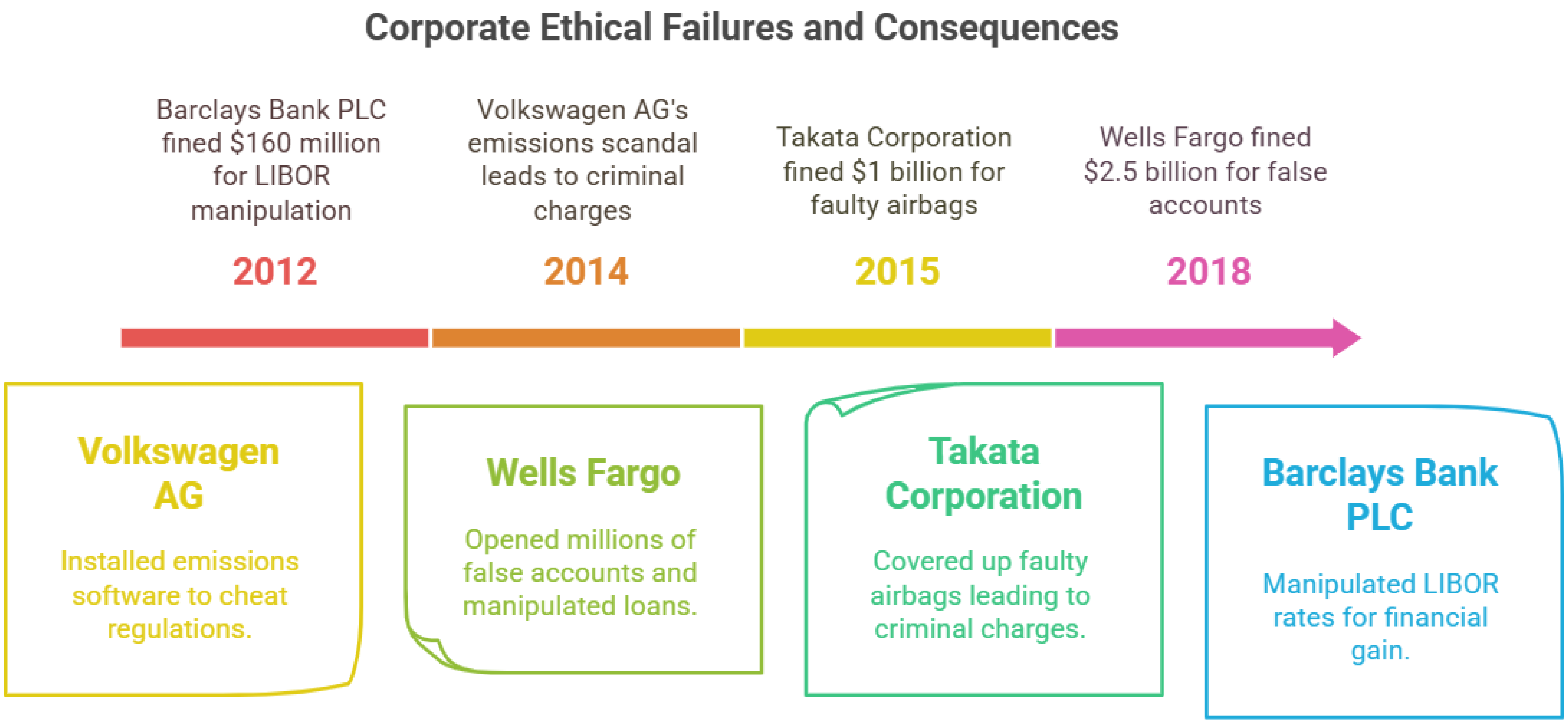
Cases of failed ethical judgment in business worrisome

- Wells Fargo, Deerfield Management, General Motors, Takata Corporation
- In many, information systems used to bury decisions from public scrutiny

Information systems raise new ethical questions because they create opportunities for:

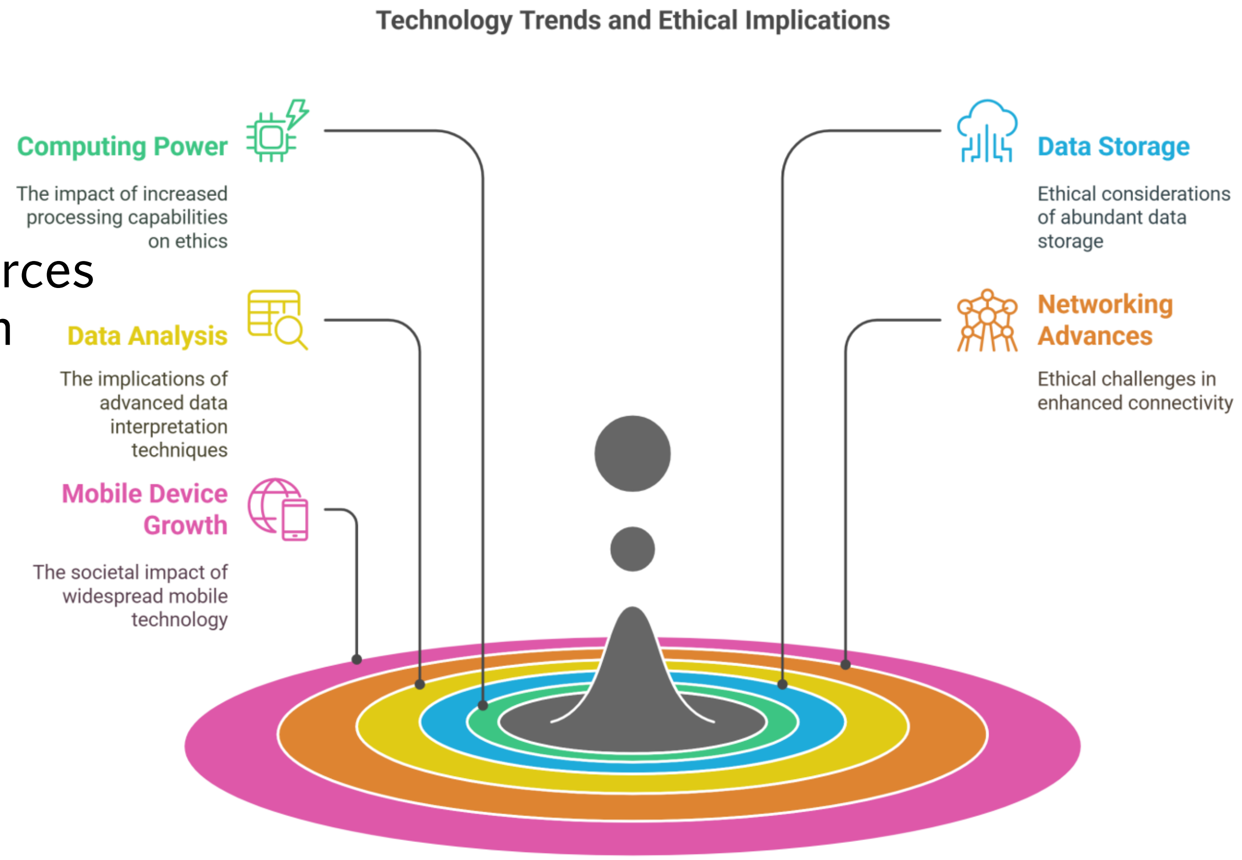
- Intense social change, threatening existing distributions of power, money, rights, and obligations
- New opportunities for crime
- New kinds of crimes

Consequence of Failures



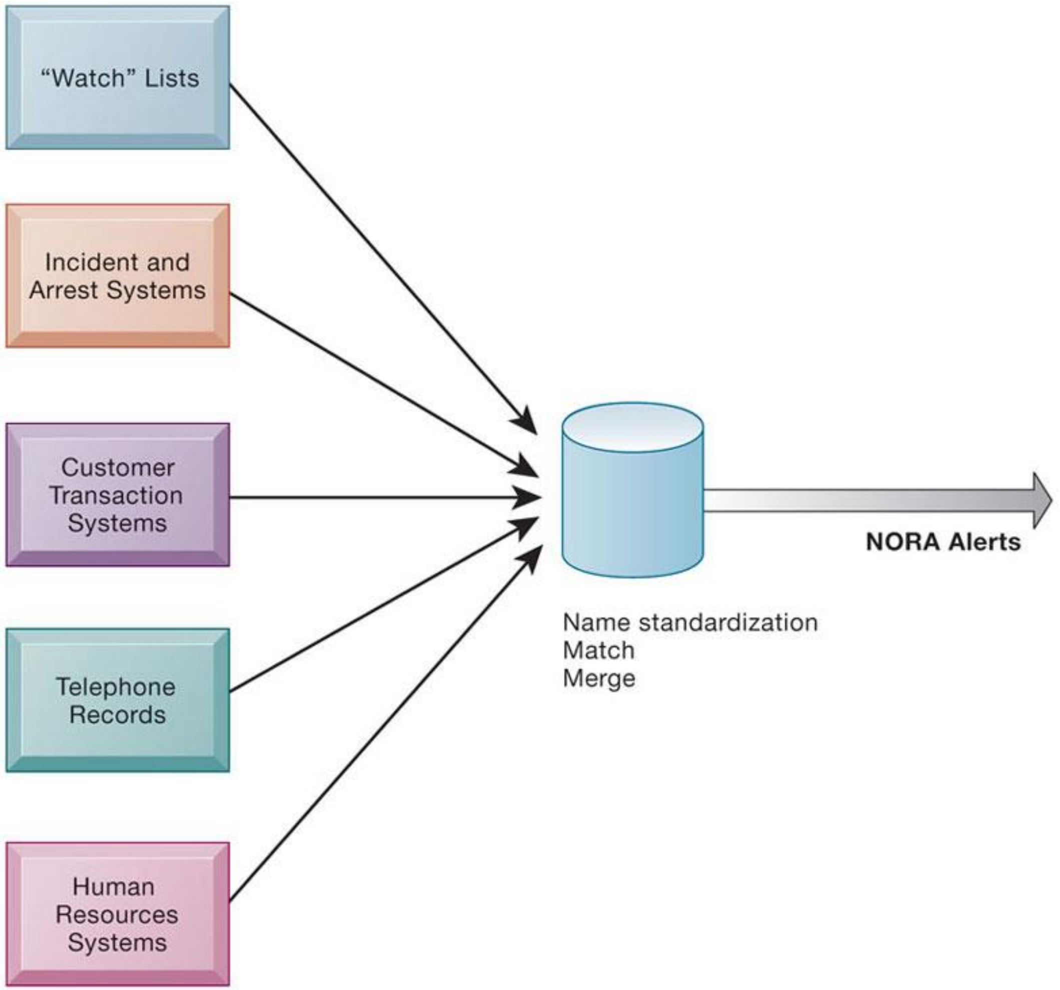
Key Technology Trends That Raise Ethical Issues

- Computing power doubles every 18 months and data storage costs rapidly decline
- Data analysis advances
  - Profiling: Combining data from multiple sources to create dossiers of detailed information on individuals
  - Making hidden connections. E.g., NORA
- Networking advances
  - Access to data becomes more difficult to control
- Mobile device growth impact
  - Possible tracking without user consent or knowledge



Example: NORA

Nonobvious relationship awareness (NORA): Combining data from multiple sources to find obscure hidden connections that might help identify criminals or terrorists




Amazing! NORA found a link–this casino employee and a high-stakes gambler lived at the same address five years ago!
Or maybe they were just old roommates. Let's not jump to conclusions.
It also found a job applicant who used to live with someone now on a watchlist...
But that was years ago. They might have no connection now.
Just because we *can* find these connections... should we *always* use them to make decisi–ons?
Security efficiency
and
Privacy taimess
NORA reveals hidden patterns... but **ethical judgment must** decide how–**and when**–to use them.

Other Examples

- Monitoring employees: Right of company to maximize productivity of workers versus workers' desire to use Internet for short personal tasks
- Facebook monitors users and sells information to advertisers and app developers

Competing Value	Description	Example
Convenience	Making tasks easier, faster, or more seamless through IS technology, often by automating or tracking behavior.	Employee monitoring software automatically tracks work hours and productivity; social media personalizes feeds for instant engagement.
Autonomy	Allowing individuals to maintain control over their choices, actions, and personal information without constant oversight.	Employees manage their own work schedules without surveillance; users choose how their personal data is shared on social media.

Balancing convenience and autonomy is a fundamental ethical dilemma in designing and managing information systems.

Ethical Analysis

Five-step process for ethical analysis

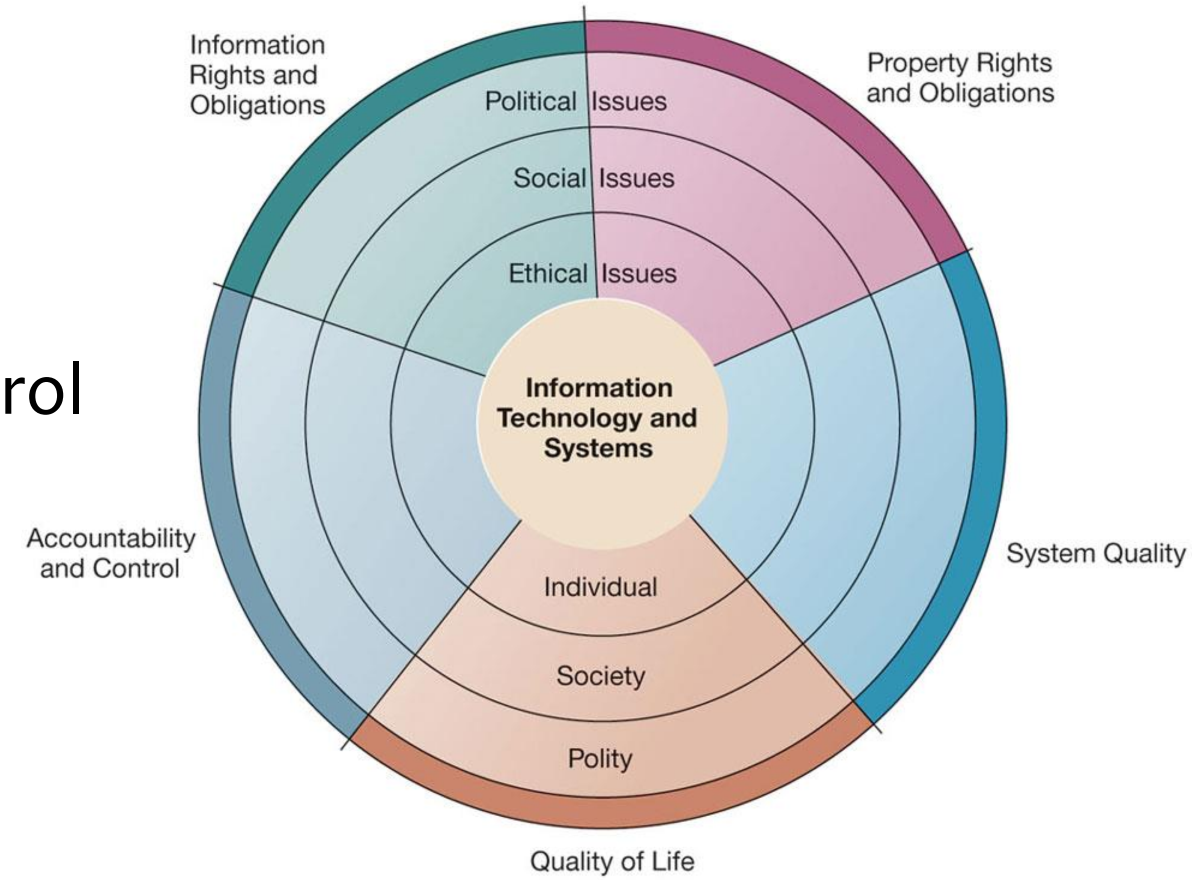
1. Identify and clearly describe the facts.
2. Define the conflict or dilemma and identify the higher-order values involved.
3. Identify the stakeholders.
4. Identify the options that you can reasonably take.
5. Identify the potential consequences of your options.

Golden Rule: Do unto others as you would have them do unto you

Slippery Slope Rule : If an action cannot be taken repeatedly, it is not right to take at all

Five Moral Dimensions of IS

- Information rights and obligations
- Property rights and obligations
- System quality
- Quality of life
- Accountability (liability) and control



Information Rights & Obligation

Privacy

- Claim of individuals to be left alone, free from surveillance or interference from other individuals, organizations, or state;
- Able to control information about yourself
- Fair information practices
  - Set of principles governing the collection and use of information
- PDPA, GDPR, APP, COPPA, HIPAA etc. implement and enforce these fair information practices in specific contexts

Property Rights: Intellectual Property

- Intellectual property
  - Tangible and intangible products of the mind created by individuals or corporations
- Protected in four main ways:
  - Copyright
  - Patents
  - Trademarks
  - Trade secret

Challenges to Intellectual Property Rights

- Digital media different from physical media
  - Ease of replication
  - Ease of transmission (networks, Internet)
  - Ease of alteration
  - Compactness
  - Difficulties in establishing uniqueness
- Digital Millennium Copyright Act (DMCA 1998)

Accountability (liability) and control

Example: Computer-Related Liability Problems

- If software fails, who is accountable?
- If seen as part of a machine that injures or harms, software producer and operator may be liable
- If seen as similar to book, difficult to hold author/publisher accountable
- If seen as a service, would this be similar to telephone systems not being liable for transmitted messages?

Computer Misuse Act

CMA 1993

- An act to criminalize unauthorized access or modification of computer material, and other computer crimes. The Act governs the investigation and prosecution of cybercrime perpetrators
  - Hacking/Hacking attempt
  - Unauthorized
  - Access/modification/use/interception/obstruction/disclosure
  - Attacks to protected computers

https://sso.agc.gov.sg/Act/CMA1993?ValidDate=20180831&ProvIds=legis

Computer Misuse Act (cont.)

PART 2 OFFENCES

- 3 - Unauthorized access to computer material
- 4 - Access with intent to commit or facilitate commission of offence
- 5 - Unauthorized modification of computer material
- 6 - Unauthorized use or interception of computer service
- 7 - Unauthorized obstruction of use of computer
- 8 - Unauthorized disclosure of access code
- 9 - Supplying, etc., personal information obtained in contravention of certain provisions
- 10 - Obtaining, etc., items for use in certain offences
- 11 - Enhancing punishment for offences involving protected computers
- 12 - Abetments and attempts punishable as offences

System Quality: Data Quality and System Errors

- What is an acceptable, technologically feasible level of system quality?
  - Flawless software is economically unfeasible
- Three principal sources of poor system performance
  - Software bugs, errors
    - E.g., case of software and sensor malfunction in Boeing 737 MAX
  - Hardware or facility failures
    - E.g., Google cloud outage in 2018 took down entire platforms – Snapchat, Spotify, and Pokemon GO
  - Poor input data quality and practices (most common source of business system failure)
    - E.g., Singapore Airlines Pricing Glitch (2014)

Quality of Life: Equity, Access, Boundaries

- Negative social consequences of systems
- Balancing power: center versus periphery
- Rapidity of change: reduced response time to competition
- Maintaining boundaries: family, work, and leisure
- Dependence and vulnerability
- Computer crime and abuse
- Equity and access
    - The digital divide
- Health risks
    - Repetitive stress injury (RSI)
    - Carpal tunnel syndrome (CTS)
    - Computer vision syndrome (CVS)
    - Technostress

Class Activity: Match it

TASK:
Match, on the next slide, each term in Column A with its corresponding description in Column B by writing the correct letter next to each number.

Class Activity: Match it

Column A	Column B
1. Competitive Advantage Paradox	A. A Singapore law that governs the collection, use, and disclosure of personal data while balancing individual privacy rights with organizational needs
2. Business-IT Alignment	B. The integration of technological change with organizational culture and behavior change, exemplified by Sharp's Yammer implementation
3. NORA (Nonobvious Relationship Awareness)	C. "Do unto others as you would have them do unto you" — a principle for evaluating whether an action is ethically appropriate
4. PDPA	D. The challenge of ensuring technology investments support business objectives, with only 25% of firms achieving proper alignment
5. Critical Information Infrastructure (CII)	E. Systems essential for continuous delivery of national services whose compromise would have debilitating effects, regulated under Singapore's Cybersecurity Act
6. Socio-technical Change	F. The situation where strategic advantages gained from information systems become increasingly temporary as innovations are quickly copied
7. The Golden Rule (in Ethical Analysis)	G. Technology that combines data from multiple sources to find hidden connections between people, raising both security benefits and privacy concerns

Class Activity: Match it

Column A	Column B (Answers)
1	F
2	D
3	G
4	A
5	E
6	B
7	C

Next

Critical IS Infrastructure – I

Aside: Internet Challenges to Privacy (not for exam)

Internet Challenges to Privacy

- Cookies
  - Identify browser and track visits to site
  - Super cookies (Flash cookies)
- Web beacons/Pixels (web bugs)
  - Tiny graphics embedded in e-mails and web pages
  - Monitor who is reading email message or visiting site
- Spyware
  - Surreptitiously installed on user's computer
  - May transmit user's keystrokes or display unwanted ads
- Google services and behavioral targeting

Internet Challenges to Privacy (cont.)

- Businesses gather transaction information and use this for other marketing purposes.
    - Web beacons/Pixels for tracking and dynamic pricing
    - App based tracking for flash sales notifications
- Opt-out vs. opt-in model
- Online industry promotes self-regulation over privacy legislation.
    - Complex/ambiguous privacy statements
    - Opt-out models selected over opt-in

Technical Solutions

- Solutions include:
  - Email encryption
  - Anonymity tools
  - Anti-spyware tools
- Overall, technical solutions have failed to protect users from being tracked from one site to another
  - Browser features
    - “Private” browsing
    - “Do not track” options

Cybersecurity Act

Cybersecurity Act (March 2018)

- An Act to require or authorize the taking of measures to prevent, manage and respond to cybersecurity threats and incidents, to regulate owners of critical information infrastructure, to regulate cybersecurity service providers, and for matters related thereto, and to make consequential or related amendments to certain other written laws.

https://sso.agc.gov.sg/Acts-Supp/9-2018/

Cybersecurity Act (cont.)

Critical information infrastructure (CII)

It refers to a computer or a computer system located wholly or partly in Singapore, necessary for the continuous delivery of an essential service, and the loss or compromise of the computer or computer system will have a debilitating effect on the availability of the essential service in Singapore

- Cybersecurity Code of Practice (CCoP 2022) for CII 2.0
  - specify the minimum protection policies that a Critical Information Infrastructure Owner (CIIO) shall implement to ensure the cybersecurity of its CII

11 critical sectors:

1. Energy
2. Water
3. Banking and finance
4. Healthcare
5. Land transport
6. Aviation
7. Maritime
8. Info-communications
9. Media
10. Security and emergency services
11. Government

https://sso.agc.gov.sg/Acts-Supp/9-2018/?ProvIds=Sc1-#Sc1-

Cybersecurity Act (cont.)

Strengthen the protection of CIIs against cyber-attacks.

The Act provides a framework for the designation of CIIs. It provides CII owners with clarity on their obligations to protect CIIs from cyber-attacks, and requires the owners to report cybersecurity incidents to CSA.

Authorise CSA to prevent and respond to cybersecurity threats and incidents.

The Act empowers the Commissioner of Cybersecurity to investigate cyber threats and incidents to determine their impact and prevent further harm. These powers are calibrated based on the severity of the threat or incident and the measures required.

Establish a framework for sharing cybersecurity information.

The Act facilitates information sharing, which is critical as timely information helps the Government and owners of computer systems identify vulnerabilities and prevent cyber incidents more effectively. The Act provides a framework for CSA to request for information, and for the protection and sharing of such information.

Establish a light-touch licensing framework for cybersecurity service providers.

Some cybersecurity services can be sensitive because the service providers performing them would know where the vulnerabilities in clients' computer systems are. Licensing cybersecurity service providers will give businesses and clients more assurance in engaging such services.